**Q1** 1 Volt is equivalent to:

(A) $\frac{\text{newton}}{\text{second}}$ (B) $\frac{\text{newton}}{\text{coulomb}}$

(C) $\frac{\text{joule}}{\text{coulomb}}$ (D) $\frac{\text{joule}}{\text{second}}$

**Q2** Two unlike charges of magnitude q are separated by a distance 5d. The potential at a point midway between them:

(A) Zero

(B) $\frac{1}{4\pi\varepsilon_0}\frac{q}{d}$

(C) $\frac{3}{2\pi\varepsilon_0}\frac{q}{d}$

(D) $\frac{5}{4\pi\varepsilon_0}\frac{q}{d}$

**Q3** Two points A and B are maintained at the potentials of 18 V and –5V respectively. What will be the magnitude of work done in moving 150 α-particles from A to B?

(A) 4 × 10–12J

(B) 8 × 10–8J

(C) 2 × 10–15J

(D) 1.1 × 10–15J

**Q4** The electric potential and electric field at a point due to a point charge are 600 V and 200 N/C respectively. Then magnitude of the point charge should be :-

(A) $3\mu C$ (B) $30\mu C$

(C) $0.2\mu C$ (D) $0.5\mu C$

**Q5** The potential at a point, due to a positive charge of $100\mu C$ at a distance of 9 m, is

(A) $10^4$ V (B) $10^5$ V

(C) $10^6$ V (D) $10^7$ V

**Q6** Two unlike charges of magnitude q are separated by a distance 2d The potential at a point midway between them is

(A) Zero

(B) $\frac{1}{4\pi\varepsilon_0}$

(C) $\frac{1}{4\pi\varepsilon_0}\cdot\frac{q}{d}$

(D) $\frac{1}{4\pi\varepsilon_0}\cdot\frac{2q}{d^2}$

**Q7** An electric charge of 20 Q is situated at the origin of X - Y co-ordinate system. The potential difference between the points. (5 a, 0) and ( -3 a, 4 a ) will be

(A) a (B) 2a

(C) Zero (D) $\frac{a}{\sqrt{2}}$

**Q8** Four charges 2C,-3 C,-4C and 5C respectively are placed at the four corners of a square. Which of the following statements is true for the point of intersection of the diagonals?

(A) $\mathbf{E}=0,\ \mathbf{V}=0$

(B) $E\neq 0, V=0$

(C) $\mathbf{E}=0,\ \mathbf{V}\neq 0$

(D) $E\neq 0, V\neq 0$

**Q9** Point charge $q_1=2\mu C$ *and* $q_2=-1\mu C$ are kept at points x = 0 and x = 6 respectively. Electrical potential will be zero at points

(A) x = 2 and x = 9

(B) x = 1 and x = 5

(C) x = 4 and x = 12

(D) x = -2 and x = 2

**Q10** Two charges of $4\mu C$ each are placed at the corners A and B of an equilateral triangle ABC of side length 0.2m in air. The electric potential at C is $\left[\frac{1}{4\pi\varepsilon_0}=9\times10^9\ \frac{N-m^2}{C^2}\right]$

(A) $9\times10^4$ V

(B) $18\times10^4$ V

(C) $36\times10^4 V$

(D) $36\times10^{-4} V$

# Answer Key

| | |
|---|---|
| Q1 | C |
| Q2 | A |
| Q3 | D |
| Q4 | C |
| Q5 | B |
| Q6 | A |
| Q7 | C |
| Q8 | B |
| Q9 | C |
| Q10 | C |

# Hints & Solutions

Note: scan the QR code to watch video solution

**Q1 Text Solution:**

One volt (1 V) is defined as the potential difference between two points in a circuit when one joule (1 J) of work is done to move one coulomb (1 C) of charge between those points.

**Video Solution:**

**Q2 Text Solution:**

The electric potential at a point due to a charge q at a distance r is given by:

$$V = \frac{1}{4\pi\epsilon_0}\frac{q}{r}$$

Since potential is a scalar quantity, we can directly add the contributions from both

$$V_{mid} = V_{+q} + V_{-q}$$

$$= \frac{1}{4\pi\epsilon_0}\frac{q}{5d/2} + \frac{1}{4\pi\epsilon_0}\frac{-q}{5d/2}$$

$$= \frac{1}{4\pi\epsilon_0}\left(\frac{q}{5d/2} - \frac{q}{5d/2}\right)$$

= 0

**Video Solution:**

**Q3 Text Solution:**

$$q_\alpha = 2e = 2 \times 1.6 \times 10^{-19} C$$

$$\Delta V = V_B - V_A = (-5) - (18) = -23 V$$

$$W_{\text{one particle}} = q \cdot \Delta V$$

$$= \left(2 \times 1.6 \times 10^{-19}\right) \times (-23)$$

$$= -7.36 \times 10^{-18} \text{ J}$$

$$W_{total} = 150 \times 7.36 \times 10^{-18}$$

$$= 1.104 \times 10^{-15} \text{ J}$$

**Video Solution:**

**Q4 Text Solution:**

$$\frac{kq}{r^2} = 200$$

$$\frac{kq}{r} = 600$$

$$\therefore r = 3$$

$$\frac{9 \times 10^9 \times q}{3} = 600$$

$$\Rightarrow q = 0.2\mu C$$

**Video Solution:**

**Q5 Text Solution:**

By using

$$V = 9 \times 10^9 \times \frac{Q}{r} = 9 \times 10^9 \times \frac{100 \times 10^{-6}}{9}$$

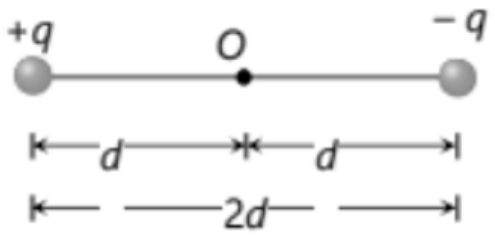
$$= 10^5 \text{ V}$$

**Video Solution:**

**Q6 Text Solution:**

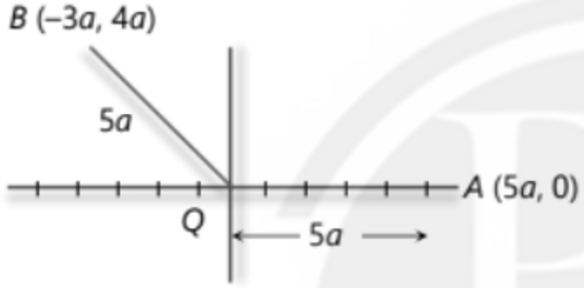
Potential at mid-point

$$O, V = \frac{kq}{d} + \frac{k(-q)}{d} = 0$$




**Video Solution:**

**Q7 Text Solution:**

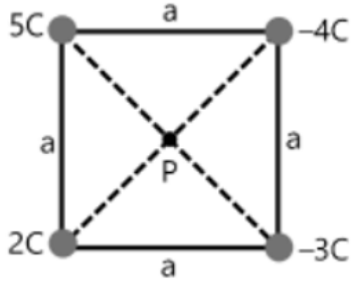



$V_A = \frac{kQ}{5a}$ and $V_B = \frac{kQ}{5a}$

$\therefore \quad V_A - V_B = 0$

**Video Solution:**

**Q8 Text Solution:**



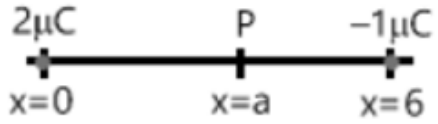

$V_p = \frac{k}{a/\sqrt{2}}\left(2-3-4+5\right) = 0$

$E_p = \sqrt{2}\frac{k}{(a/\sqrt{2})^2} \times 8 \neq 0$

**Video Solution:**

**Q9 Text Solution:**

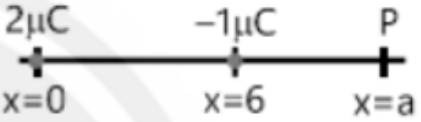
Case-I




$V_p = \frac{k\times 2}{a} - \frac{k\times 1}{6-a} = 0$

$\Rightarrow a = 4\ \text{m}$

Case-II

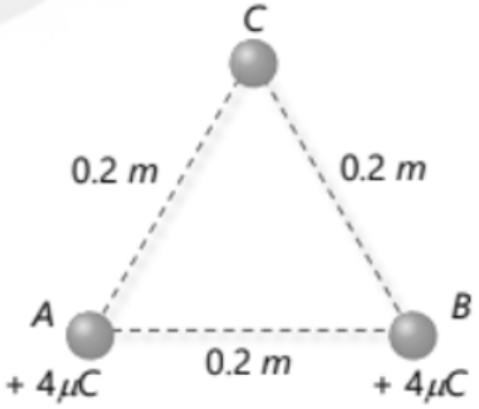



$V_P = \frac{k\times 2}{a} - \frac{k\times 1}{6+a} = 0$

$\Rightarrow a = 12\ \text{m}$

**Video Solution:**

**Q10 Text Solution:**




$Potential at\ C = \left(9\times 10^9 \times \frac{4\times 10^{-6}}{0.2}\right)\times 2$

$= 36 \times 10^4 V$

**Video Solution:**






[Android App](#) | [iOS App](#) | [PW Website](#)